ECET 250E-Linear Circuit 1
Problemset 2-Chapter 2- Basic Laws
Solution

Name: _____ Marks: _____

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1. Chapter 2, Problem 2.

Find the hot resistance of a light bulb rated 60W, 120 V.

Solution:

$p = v^2/R \quad \rightarrow \quad \mathbf{R} = v^2/p = 120^2/60 = 14400/60 = \mathbf{240\ ohms}$

2. Chapter 2, Problem 4.

(a) Calculate current *i* in Fig. 2.68 when the switch is in position 1.

(b) Find the current when the switch is in position 2.

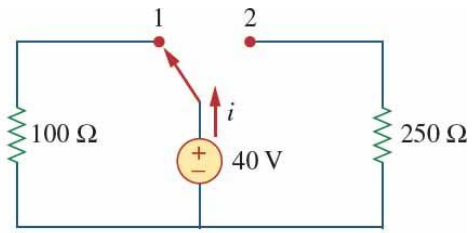


Figure 2.68
For Prob. 2.4.

Solution:

(a) Position 1: i = 40/100 = 0.400 A = **400 mA**

(b) Position 2: i = 40/250 = 0.160 A = **160 mA**

3. Chapter 2, Problem 9.

Find i_1, i_2, and i_3 in Fig. 2.73.

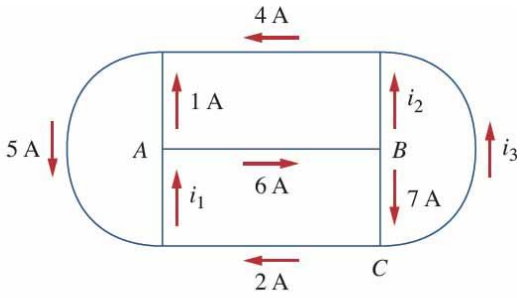


Figure 2.73
For Prob. 2.9.

Solution:

At A, $-1-6+i_1 = 0$ or $i_1 = (1+6) =$ **7 A**

At B, $6-i_2-7 = 0$ or $i_2 = 6-7 =$ **–1 A**

At C, $-2-i_3+7 = 0$ or $i_3 = 7-2 =$ **5 A**

4. Chapter 2, Problem 10.

Determine i_1 and i_2 in the circuit of Fig. 2.74.

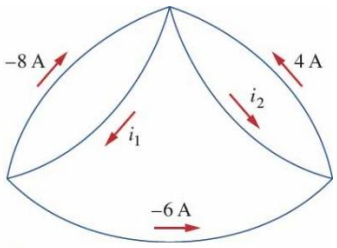
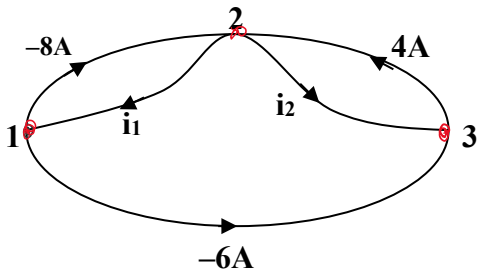


Figure 2.74
For Prob. 2.10.

Solution:



At node **1**, $-(-8)+i_1-(-6) = 0$ or $i_1 = -8-6 =$ **–14 A**

At node **2**, $(-8) - i_1 - i_2 + 4 = 0$ or $i_2 = -8 - i_1 + 4 = -8 - (\mathbf{-14}) + 4 = \mathbf{10\ A}$

5. Chapter 2, Problem 11.

In the circuit of Fig. 2.75, calculate V_1 and V_2.

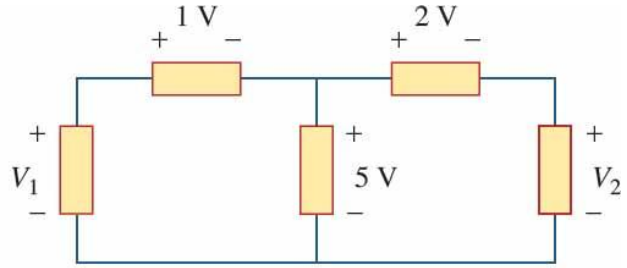


Figure 2.75
For Prob. 2.11.

Solution:

$$-V_1 + 1 + 5 = 0 \longrightarrow V_1 = \underline{6\ \text{V}}$$

$$-5 + 2 + V_2 = 0 \longrightarrow V_2 = \underline{3\ \text{V}}$$

6. Chapter 2, Problem 13.

For the circuit in Fig. 2.77, use KCL to find the branch currents I_1 to I_4.

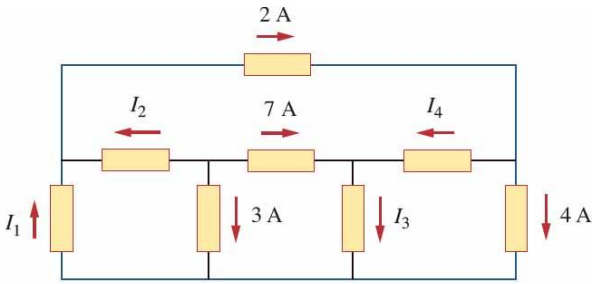
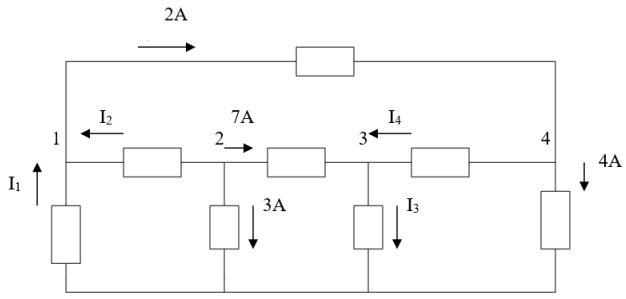


Figure 2.77
For Prob. 2.13.

Solution:



At node 2,

$$-3 - 7 - I_2 = 0 \longrightarrow \quad I_2 = -10\ \text{A}$$

At node 1,

$I_1 + I_2 = 2A \longrightarrow I_1 = 2 - (-10) = 12A$

At node 4,

$$2 = I_4 + 4 \quad \longrightarrow \quad I_4 = 2 - 4 = -2A$$

At node 3,

$$7 + I_4 = I_3 \quad \longrightarrow \quad I_3 = 7 - 2 = 5A$$

Hence,

$I_1 = 12$ A, $I_2 = -10$ A, $I_3 = 5$ A, $I_4 = -2$ A

7. Chapter 2, Problem 14.

Given the circuit in Fig. 2.78, use KVL to find the branch voltages V_1 to V_4.

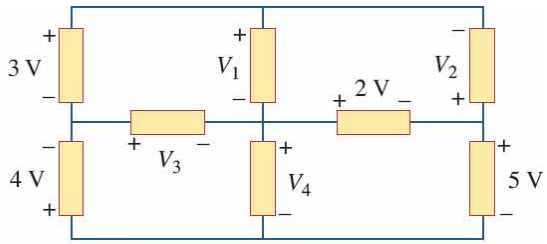
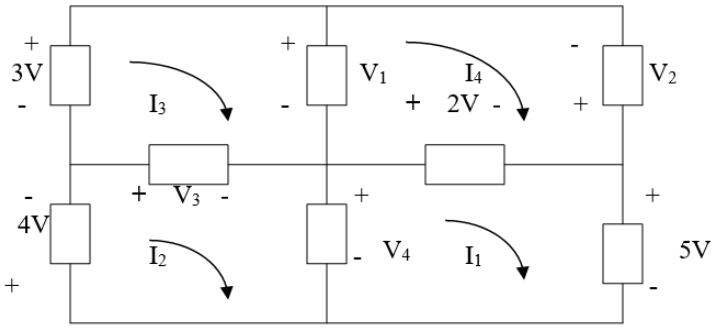


Figure 2.78
For Prob. 2.14.

Solution:



For mesh 1,

Lower right loop: $-V_4 + 2 + 5 = 0 \quad \longrightarrow \quad V_4 = 7V$

For mesh 2,

Lower left loop: $+4+V_3+V_4=0 \longrightarrow V_3=-4-7=-11V$

For mesh 3,

Upper left loop: $-3+V_1-V_3=0 \longrightarrow V_1=V_3+3=-8V$

For mesh 4,

Upper right loop: $-V_1-V_2-2=0 \longrightarrow V_2=-V_1-2=6V$

Thus,

$V_1=-8V, \quad V_2=6V, \quad V_3=-11V, \quad V_4=7V$

8. Chapter 2, Problem 15.

Calculate v and i_x in the circuit of Fig. 2.79.

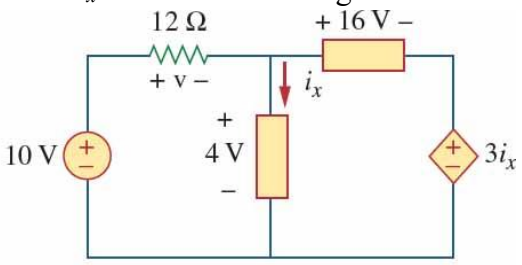


Figure 2.79
For Prob. 2.15.

Solution:

For loop 1, Left loop: $-10 + v + 4 = 0, \quad v = \mathbf{6\ V}$

For loop 2, Right loop: $-4 + 16 + 3i_x = 0, \quad i_x = \mathbf{-4\ A}$

9. Chapter 2, Problem 16.

Determine V_o in the circuit in Fig. 2.80.

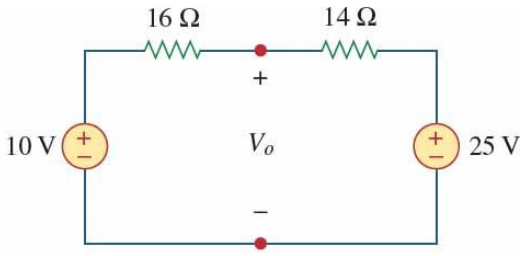


Figure 2.80
For Prob. 2.16.

Solution:

Assume a clockwise loop current I.

Apply KVL,

$-10 + (16+14)\,I + 25 = 0$ or $30I = 10-25$ $\quad I = -15/30 = -0.5\ \text{A} = -500\ \text{mA}$

Also,

$-10 + 16I + V_o = 0$ or $V_o = 10 - 16(-0.5) = 10+8 =$ **18 V**

10. Chapter 2, Problem 18.

Find I and V_{ab} in the circuit of Fig. 2.82.

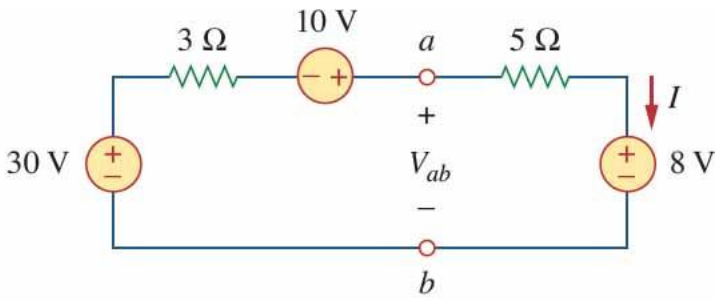


Figure 2.82
For Prob. 2.18.

Solution:

Applying KVL,

$-30 - 10 + 8 + I(3+5) = 0$

$8I = 32 \longrightarrow I =$ **4A**

$-V_{ab} + 5I + 8 = 0 \longrightarrow Vab = 5(4) + 8 =$ **28 V**

11. Chapter 2, Problem 19.

From the circuit in Fig. 2.83, find I, the power dissipated by the resistor, and the power supplied by each source.

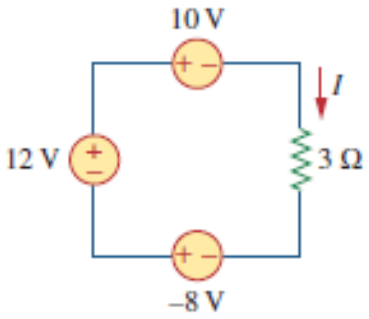


Figure 2.83
For Prob. 2.19.

Solution:

Applying KVL around the loop, we obtain

$$-(-8) - 12 + 10 + 3i = 0 \quad \rightarrow \quad 8 - 12 + 10 + 3I = 0 \quad \rightarrow \quad \mathbf{i = -2A}$$

Power dissipated by the resistor:

$$p_{3\Omega} = i^2R = 4(3) = \mathbf{12W}$$

Power supplied by the sources:

Power supplied by the 12 V source: $P_{12v} = 12I = \mathbf{-24\ W}$
Power supplied by the 10 V source: $P_{10v} = -10I = \mathbf{20\ W}$
Power supplied by the −8 V source: $P_{-8v} = (-8)I = \mathbf{16\ W}$

The 12 V source absorbs 24 W. The other two sources supply 20 W and 16 W.
Power balance: 20 + 16 = 24 + 12 = 36 W